

Department of Electronics & Communications Engineering

ECNG410401 - ASIC Design Using CAD (Fall 2026)

Dr. Ahmed Saeed

FPGA-to-ASIC Implementation of a RISC Processor

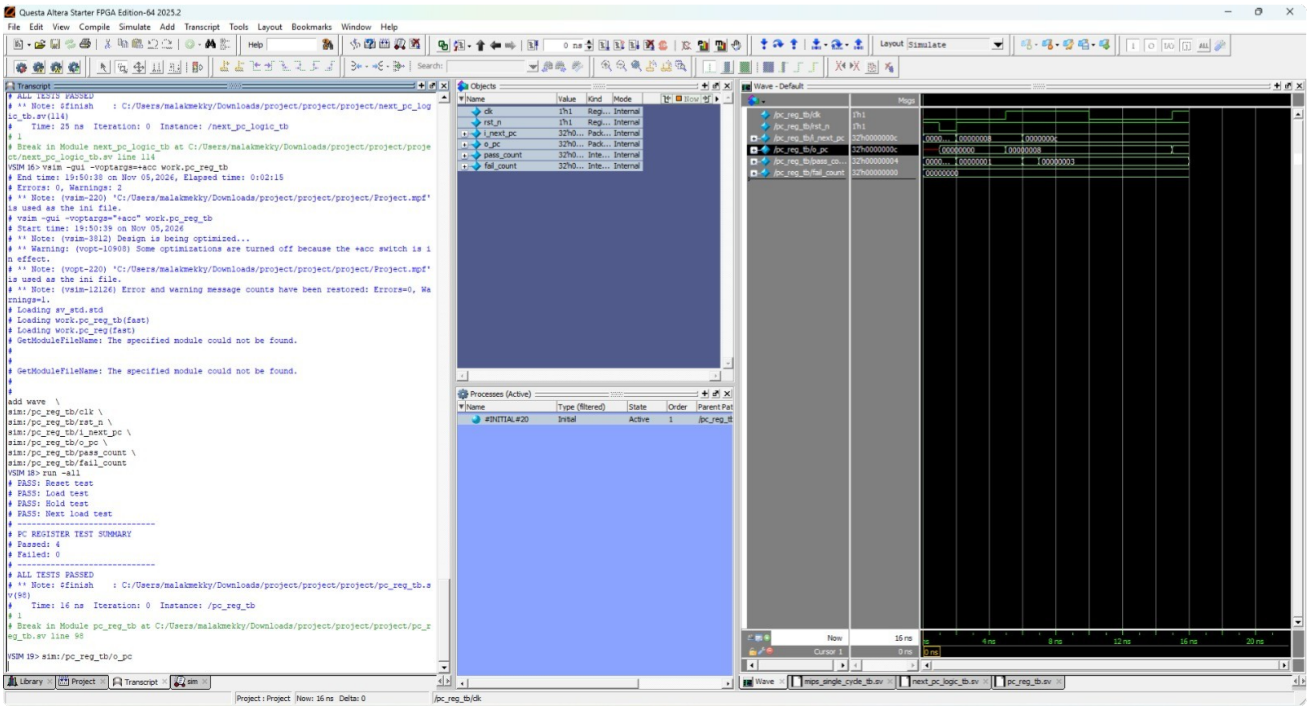
Phase I Report

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I. Introduction

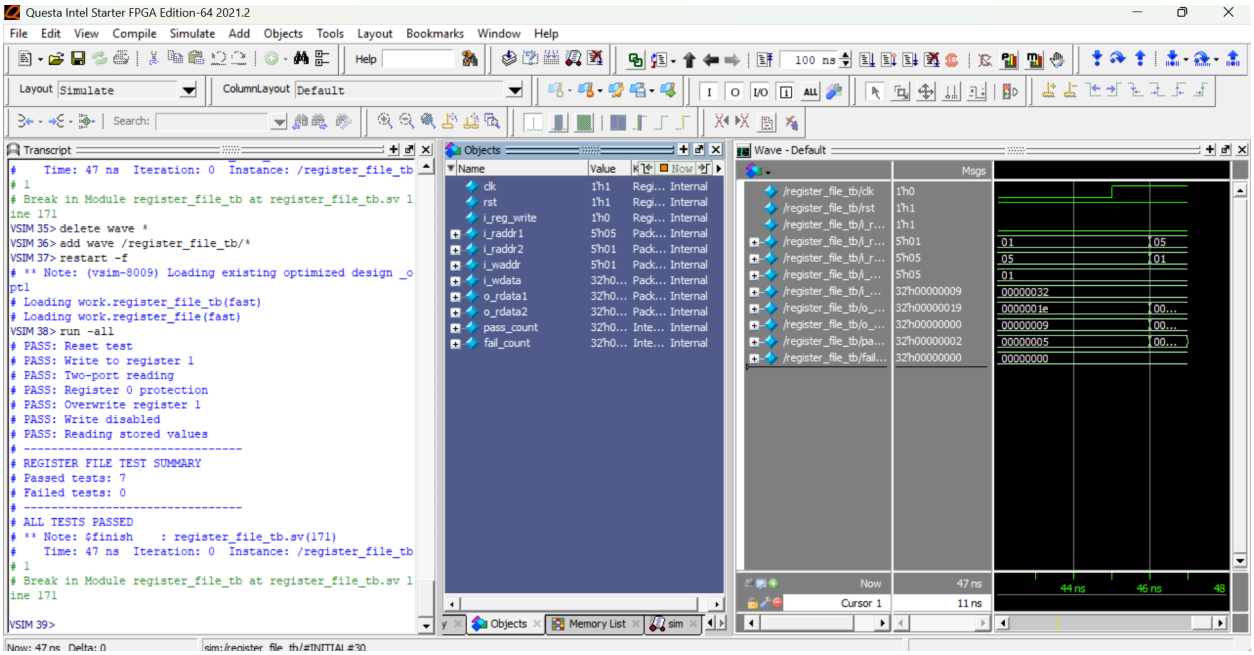
This report covers the three Phase I milestones. The RTL and self-checking testbench for each block. The integration into mips_single_cycle. And the synthesis, place and route, timing, power analysis, and the board demonstration.

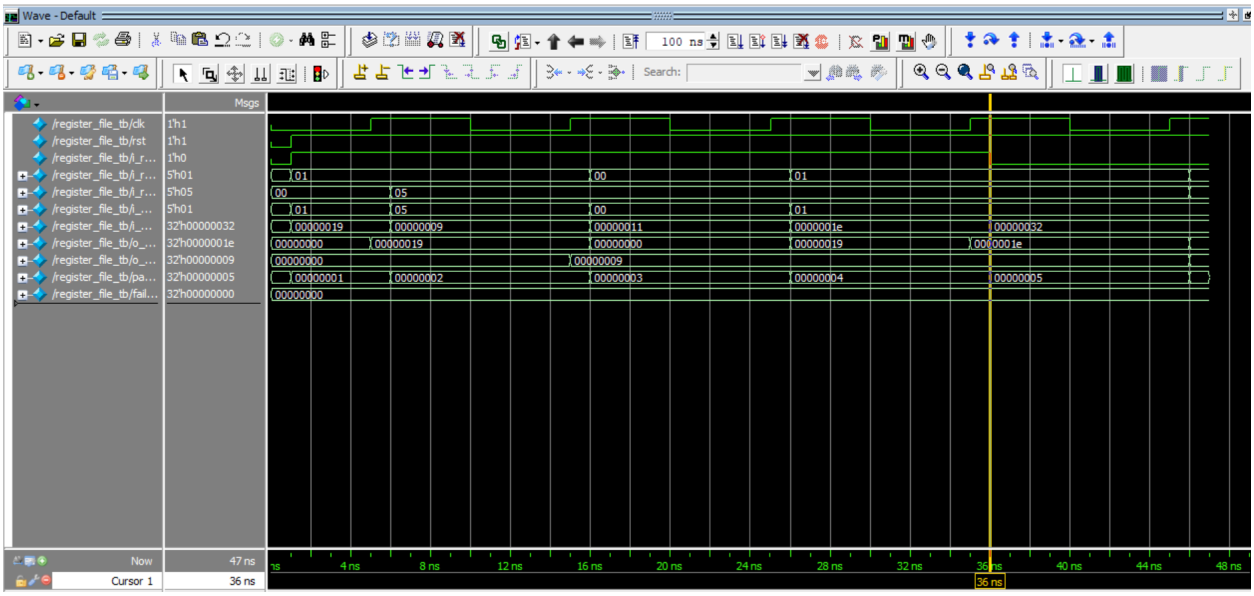
II. Program counter TB



The testbench checks reset, loading a new address, holding the value between clock edges, and loading the next address. All 4 tests pass. In the waveform, o_pc is 0 during reset then becomes 8 then 12. The input (i_next_pc) changes before the clock edge, but the PC only takes the new value on the next rising edge. This is expected for an edge-triggered register with asynchronous reset.

III. Register File TB

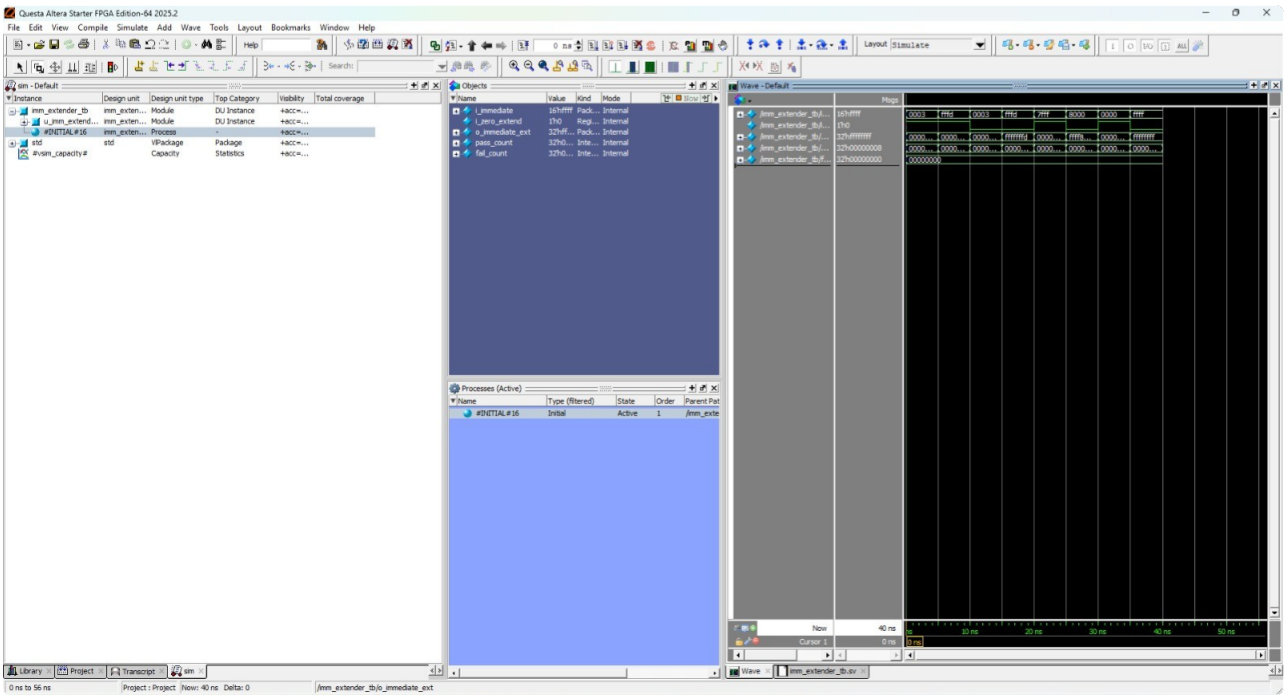


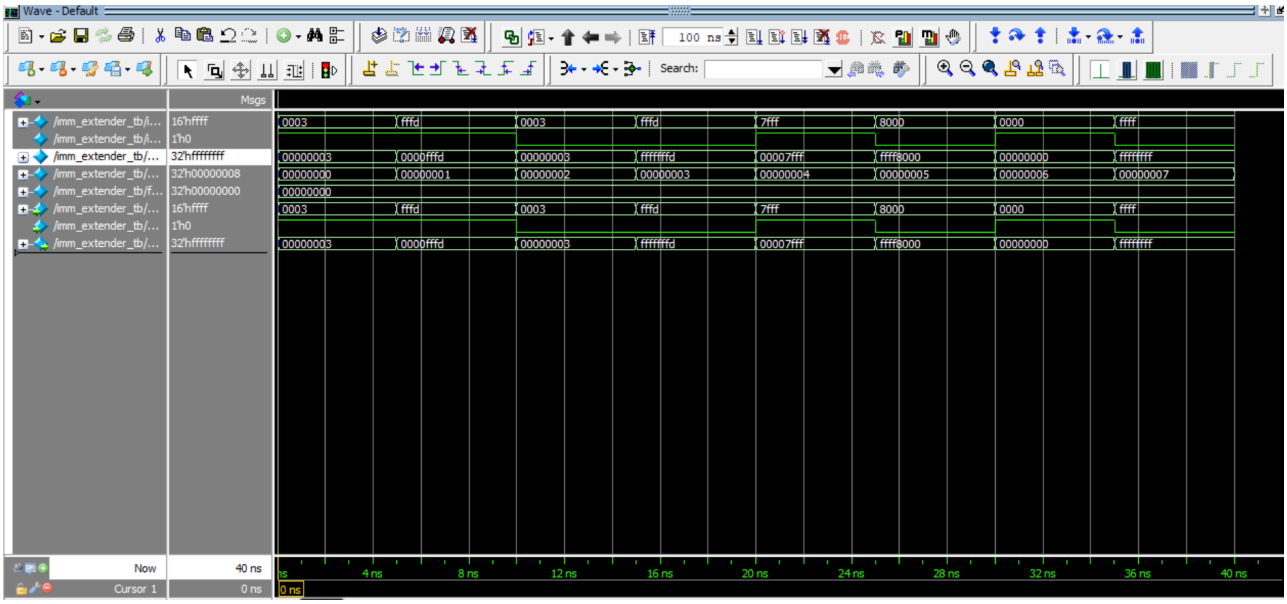


The testbench checks reset, write, reading 2 ports at once, R0 staying at 0, overwriting a register, writes disabled, and reading stored values. All tests pass. The waveform shows R0 still reading 0 after writing 17 to address 0.

IV. Immediate Extender TB

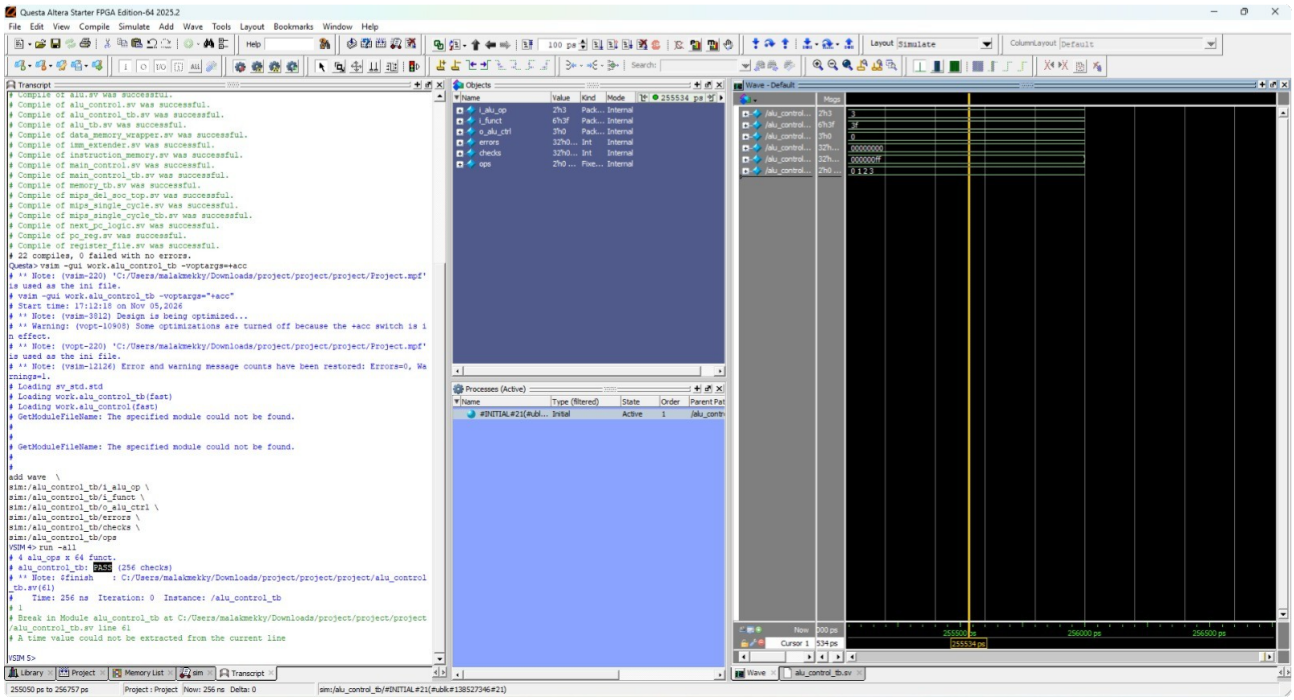
All 8 tests pass. The waveform shows the expected extensions. 0xFFFD sign extends to 0xFFFFFFFD, 0x8000 to 0xFFFF8000, and 0x7FFF stays 0x00007FFF. i_zero_extend = 1 so the upper 16 bits are always 0.





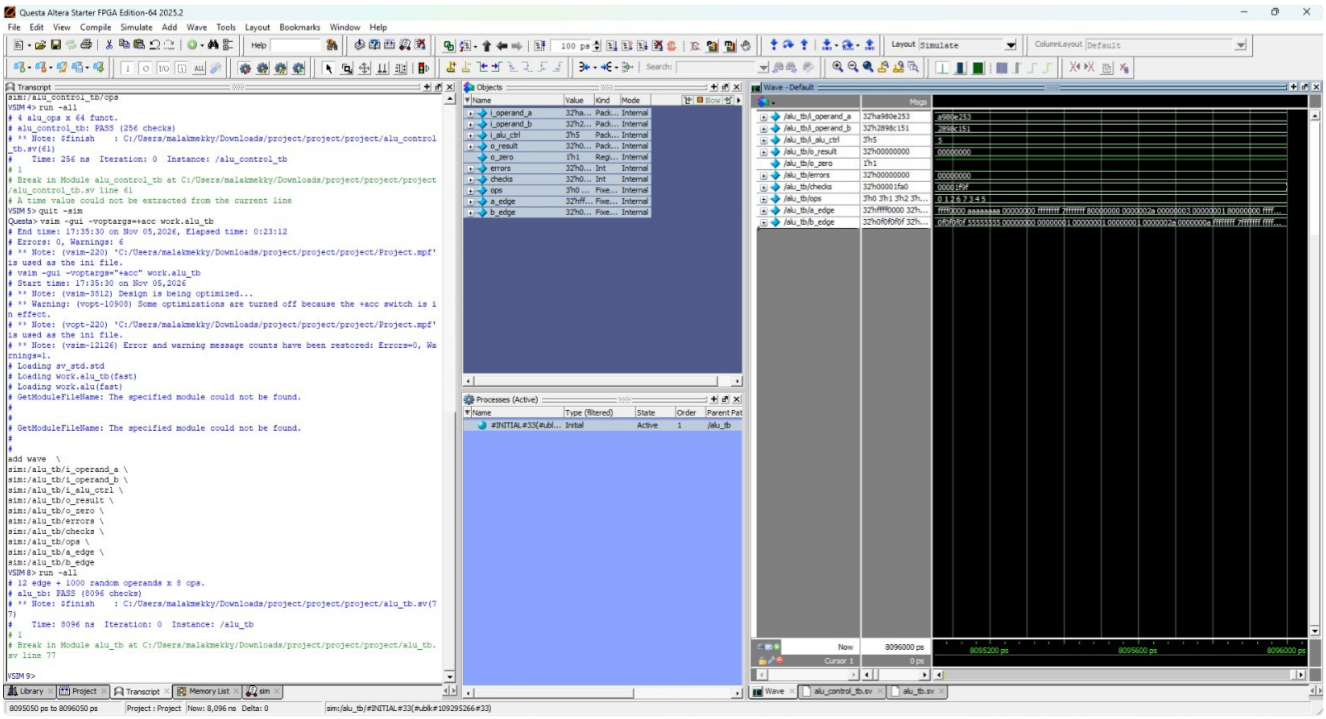
V. ALU Control TB

The testbench applies all 4 ALUOp values with all 64 funct codes and compares the output with the ALU Control table. All the 256 checks pass.

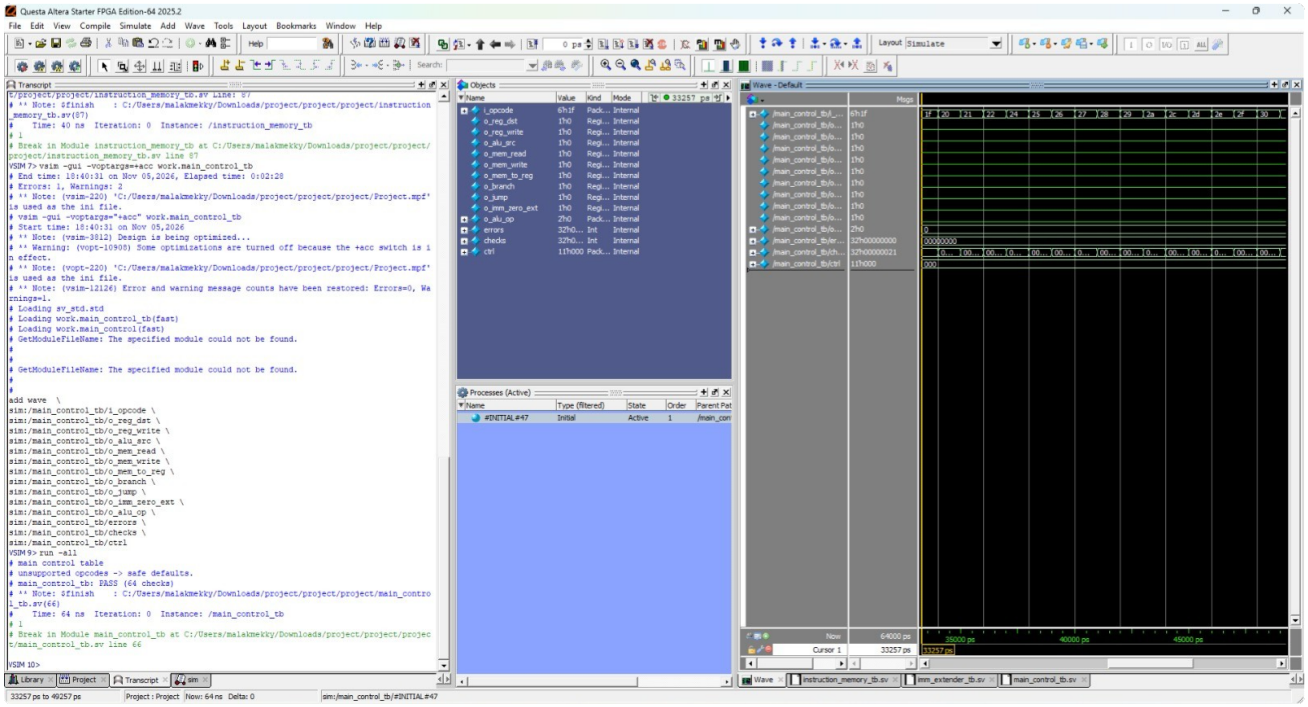


VI. ALU TB

This testbench runs 12 edge-case operand pairs plus 1000 random pairs through all 8 control codes and all the checks pass. The edge cases include values like 0x7FFFFFFF and 0x80000000 that check that slt (set if less than) compares signed numbers.



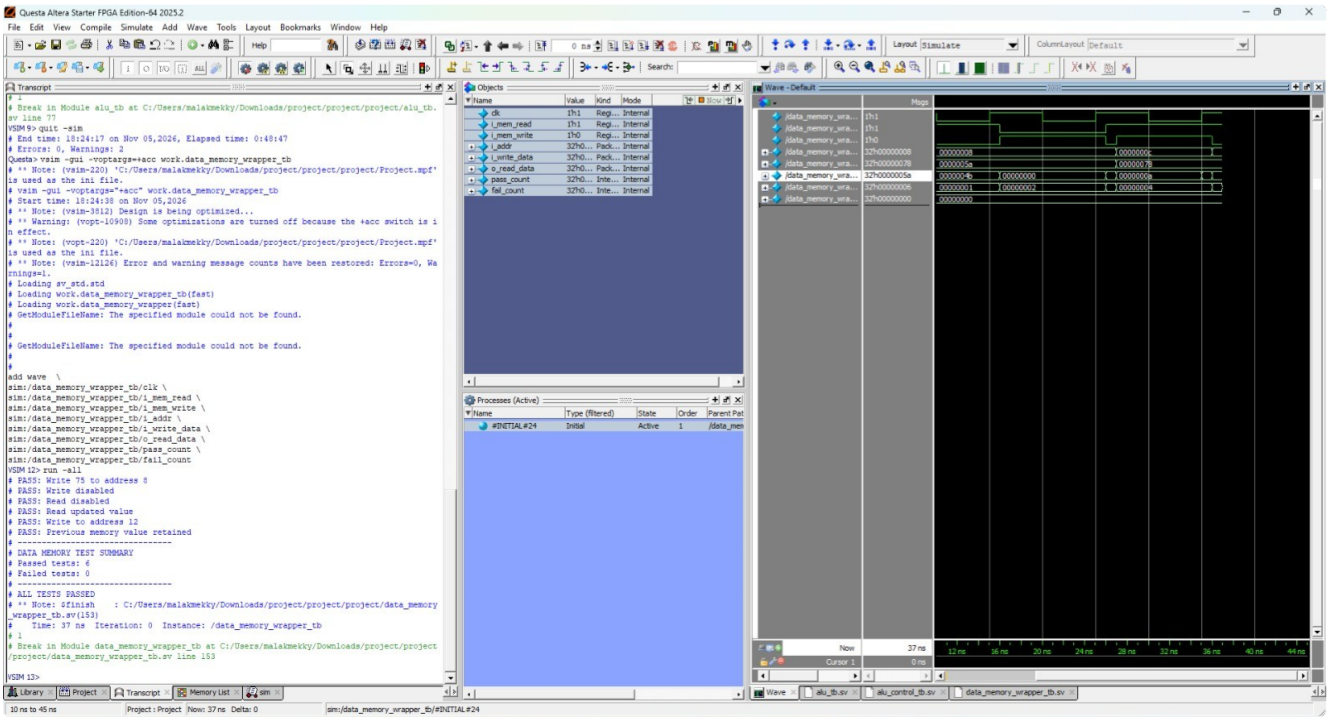
VII. Main Control TB



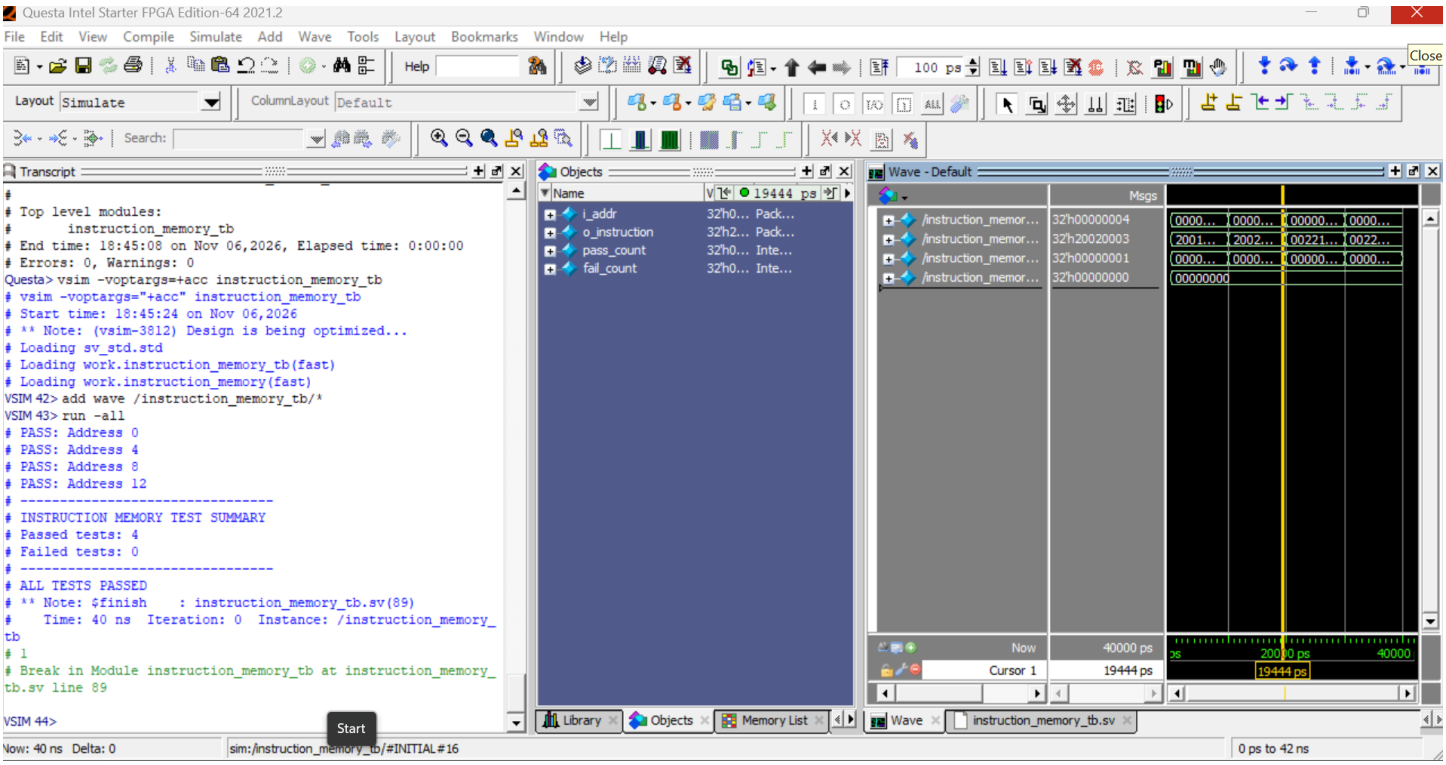
All 7 rows of the main control table are checked and the 57 unsupported opcodes. They leave every output at a default of 0. All 64 checks pass. The ctrl in the waveform stays 000 for the unsupported opcodes. No output is ever unknown which would mean a latch or missing assignment.

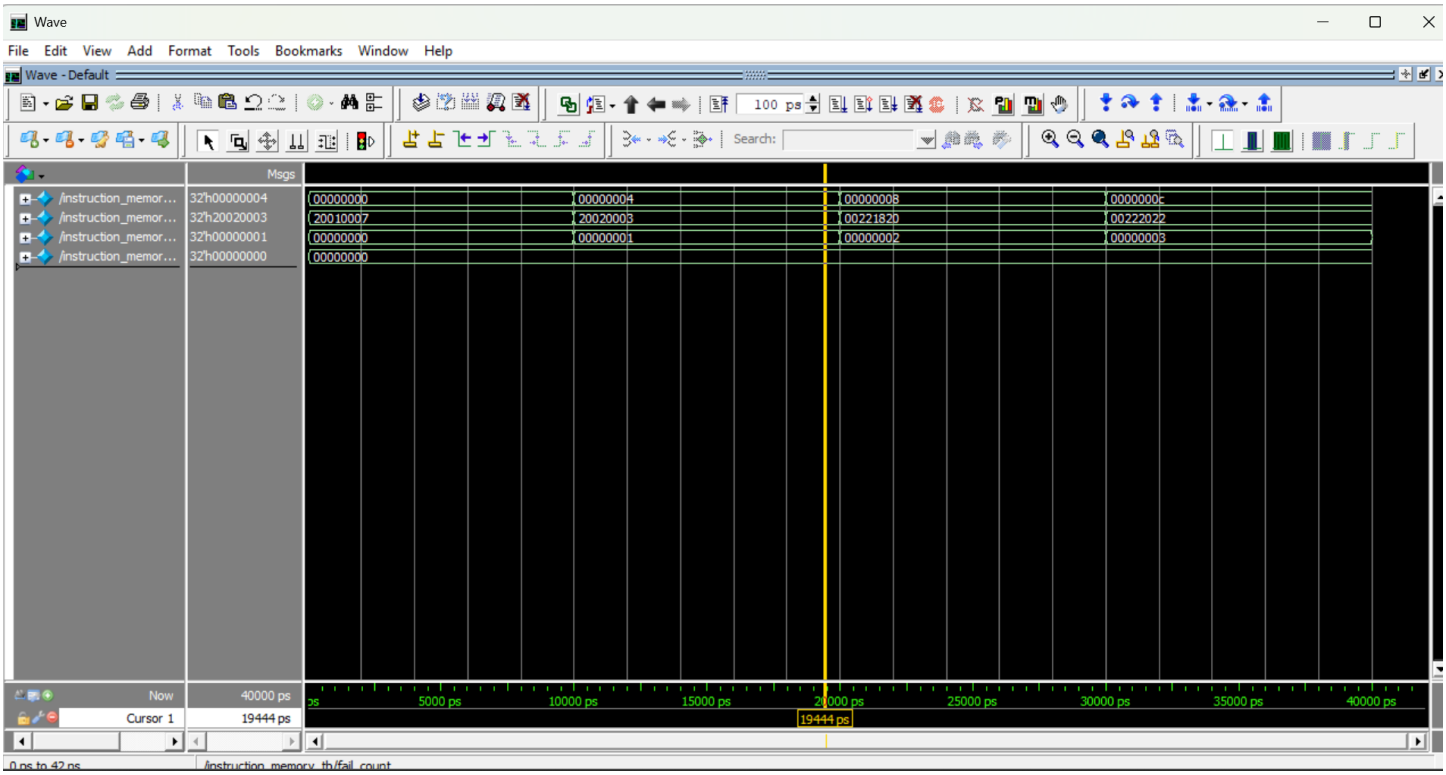
VIII. Data Memory TB

The testbench writes 75 to address 8, checks that writes are ignored when i_mem_write = 0, that output is 0 when i_mem_read = 0, that the updated value reads back, and that writing address 12 leaves address 8 the same. All 6 tests pass. It matches a synchronous-write, combinational-read memory.



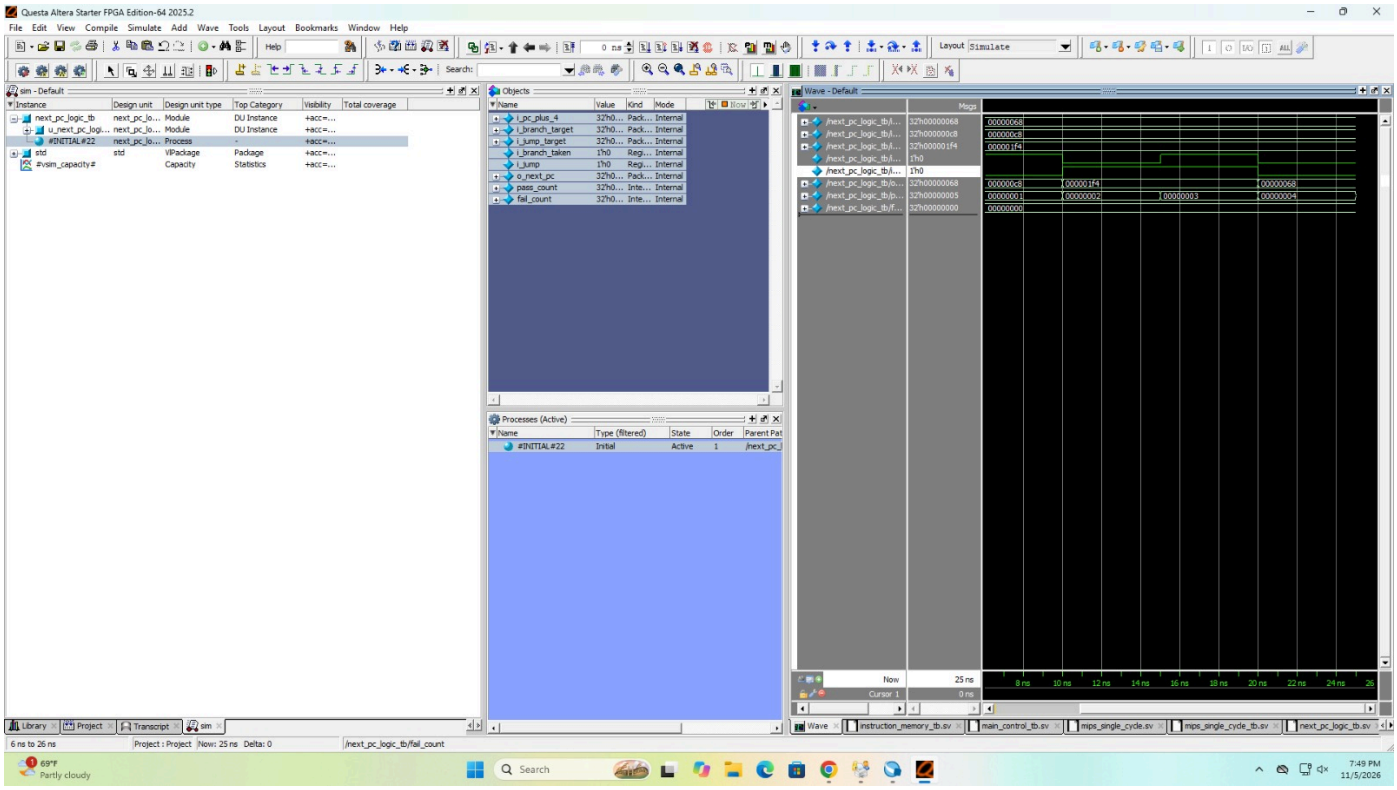
IX. Instruction Memory TB





The testbench loads test_alu.hex and reads the first four word addresses. All 4 tests pass. The waveform shows o_instruction returning the first four instructions in Test 1 in Appendix A. Each new value appears as soon as i_addr changes, no clock, which shows that the memory is read combinationally.

X. Next PC Logic TB

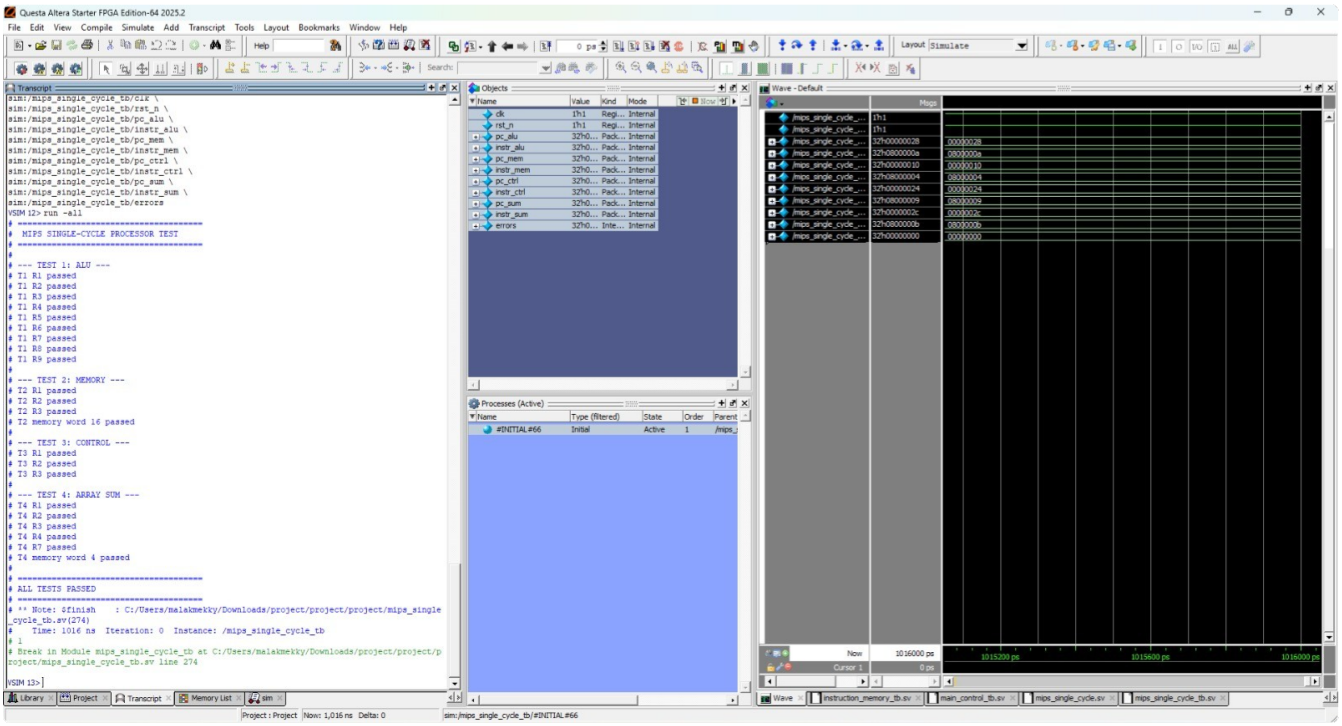


The testbench checks normal execution (PC + 4 = 0x68), a taken branch (0xC8), a jump (0x1F4), both at the same time, and neither. All 5 tests pass. When i_branch_taken and i_jump are both 1 then the output is the jump target so jump has priority.

The branch target and jump target equations don't have their own testbench and they're verified at the processor level. We only reach the final correct PC if both targets are computed correctly.

XI. Processor Integration

It passes all four of the Appendix A test programs. The mips_cycle_single_tb creates 4 processors and each one loads a .hex file. It resets, runs about 100 clock cycles, then reads the register file and data memory directly and compares them with the expected values.

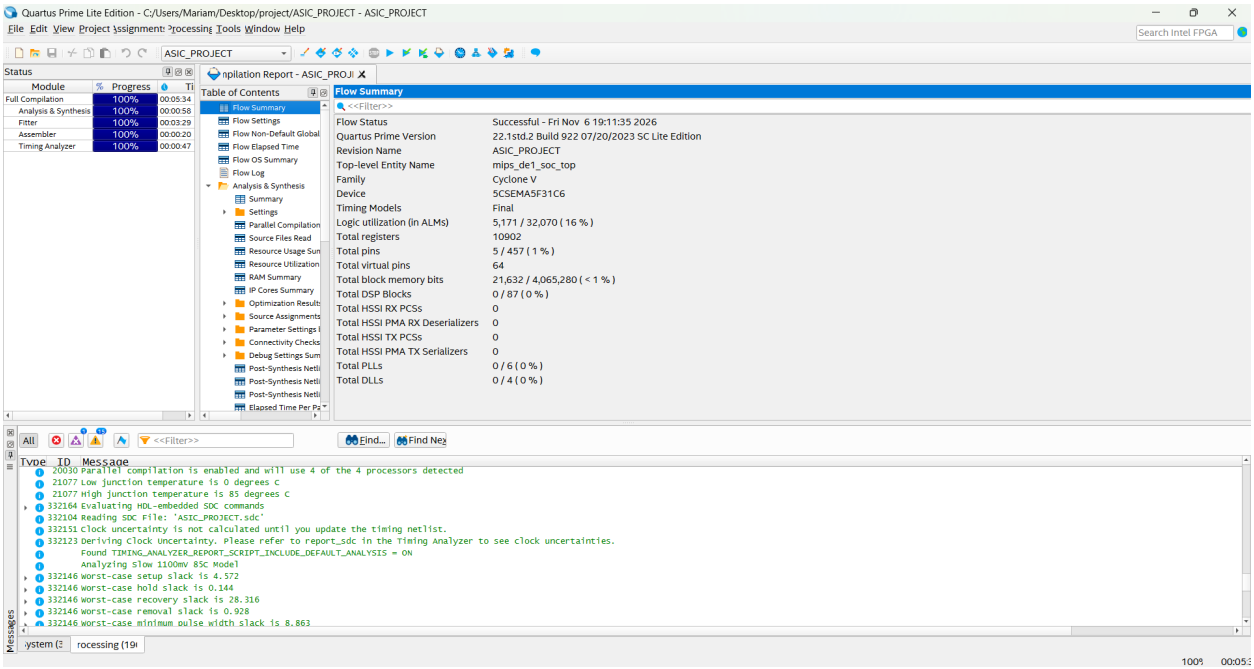


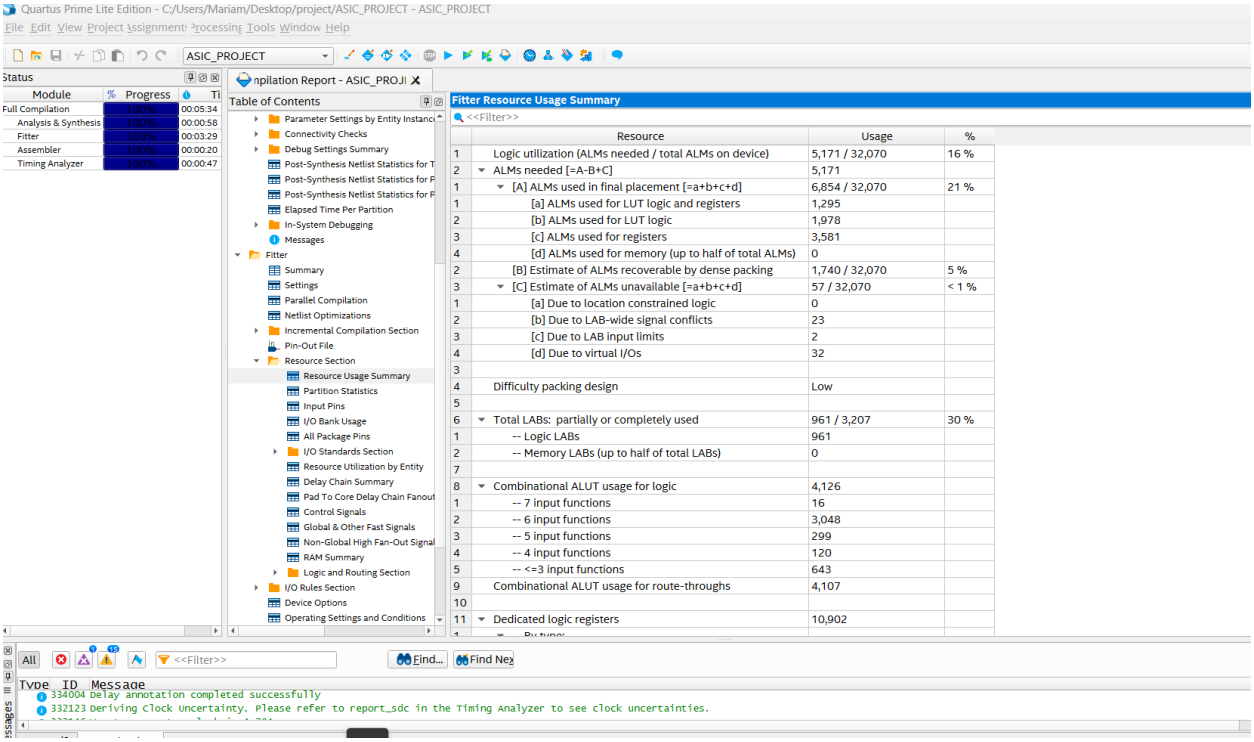
The waveform shows each processor instance's final PC on its j_end loop (0x28, 0x10, 0x24, and 0x2C) which match the addresses in the appendix. All four programs finish and stop where expected.

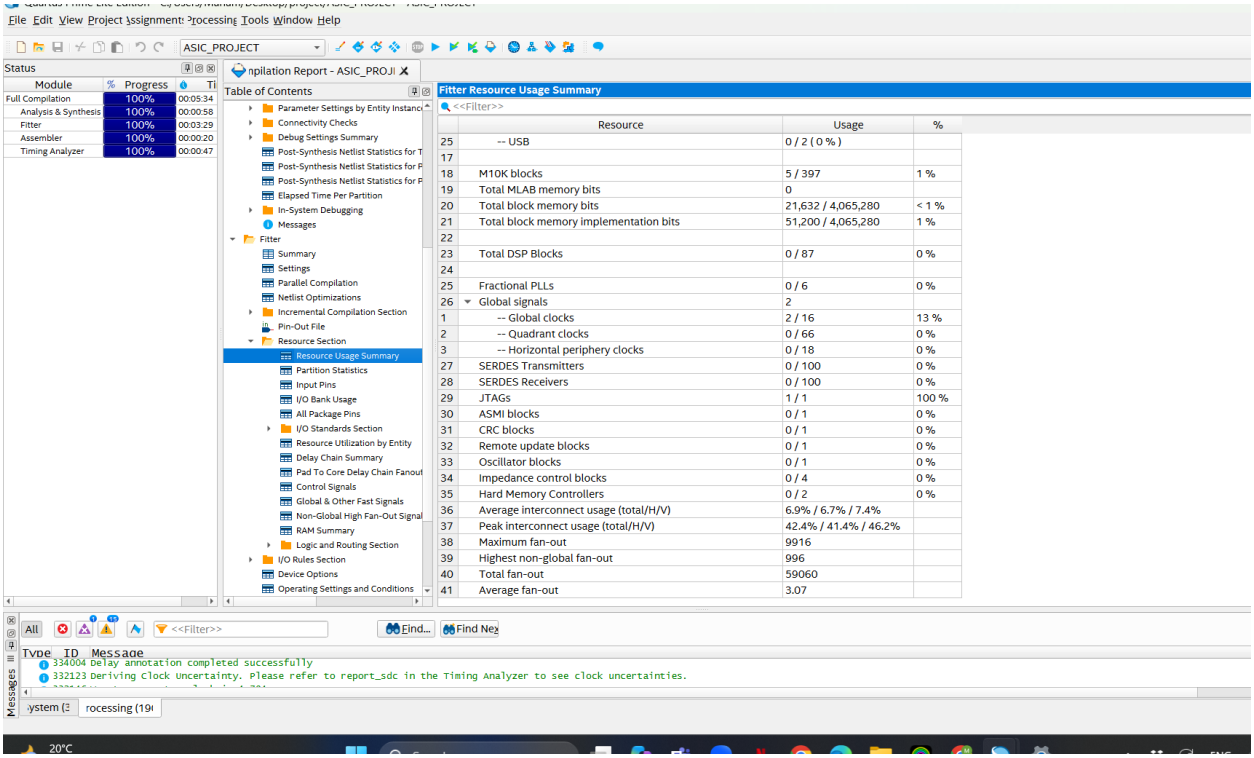
XII. FPGA Implementation

The design compiled with no errors.

1. Resource Usage Summary

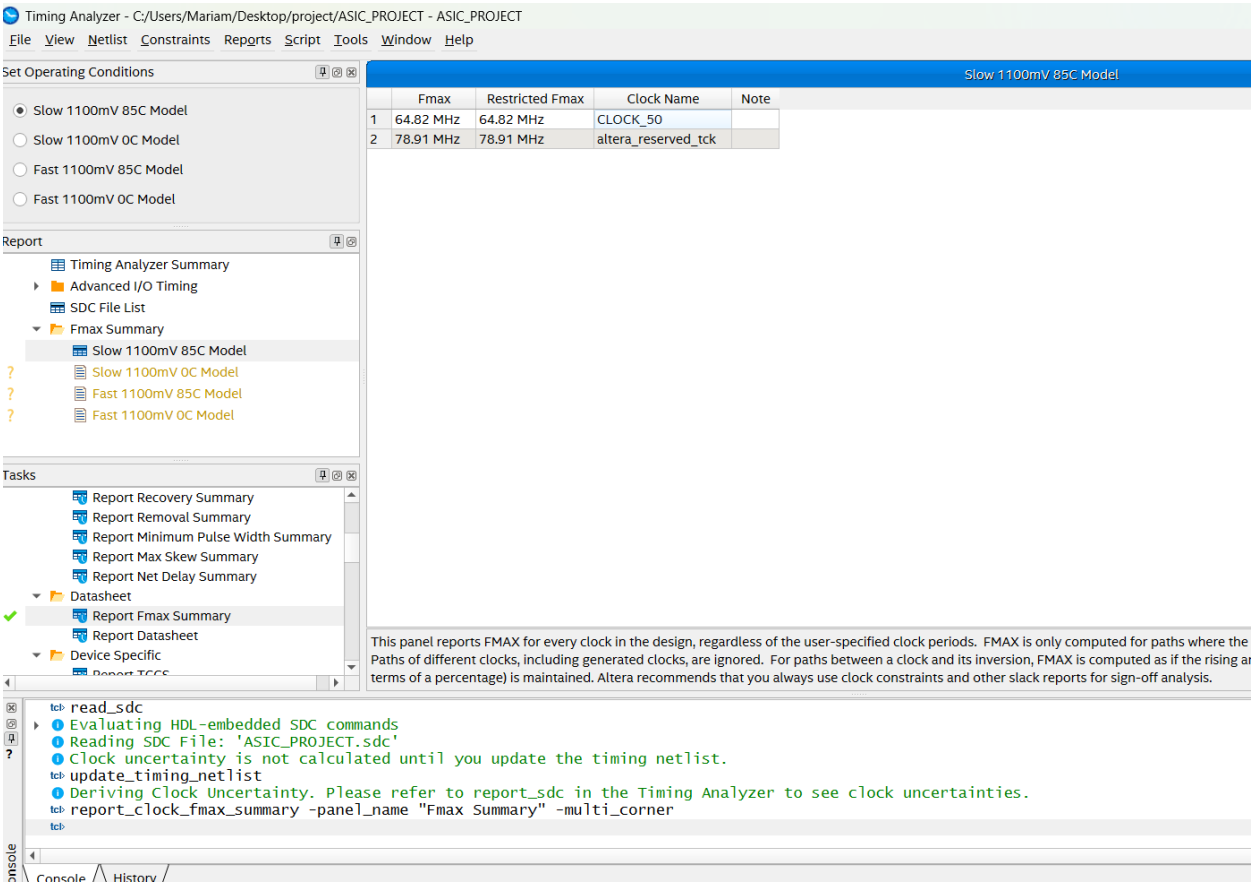




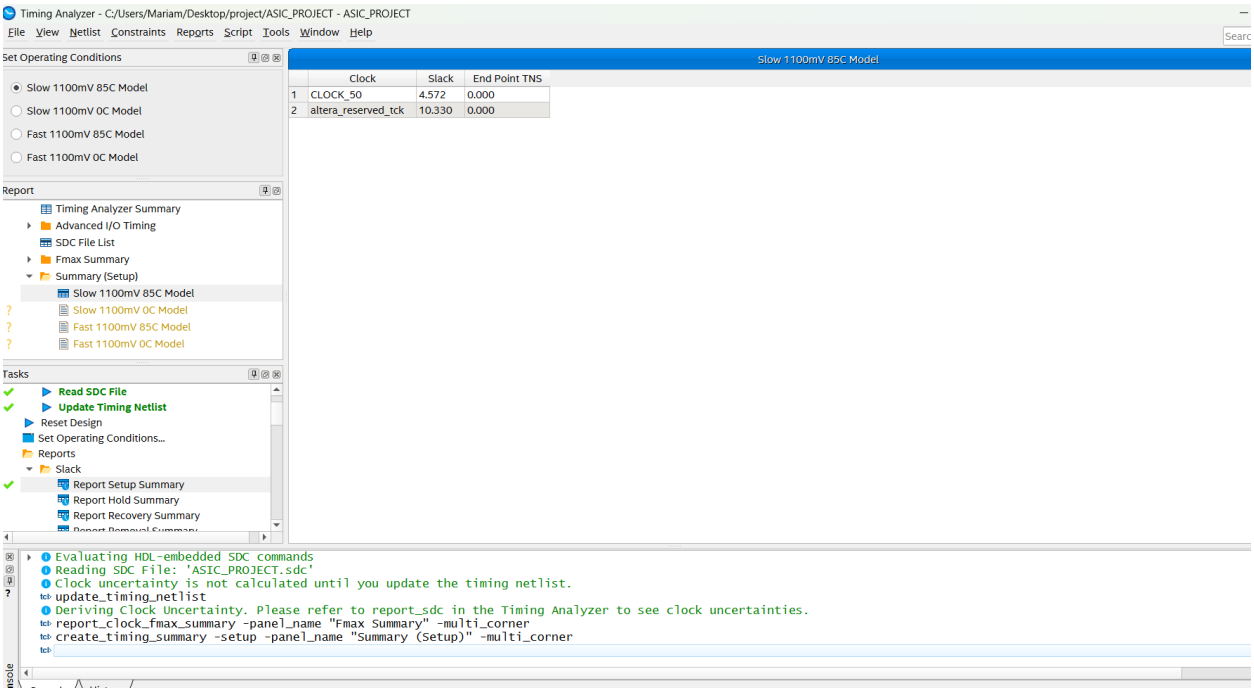


We use only 16% of the FPGA's logic (5171 of 32070 ALMs), 10902 registers, and no DSP blocks. Most of the registers come from the data memory. It's 8192 flip flops because it's read combinationally. RAM blocks only support clocked reads. The results match what's expected of a single cycle design which is no multiplier (no DSP blocks), small control logic, and a large but simple memory.

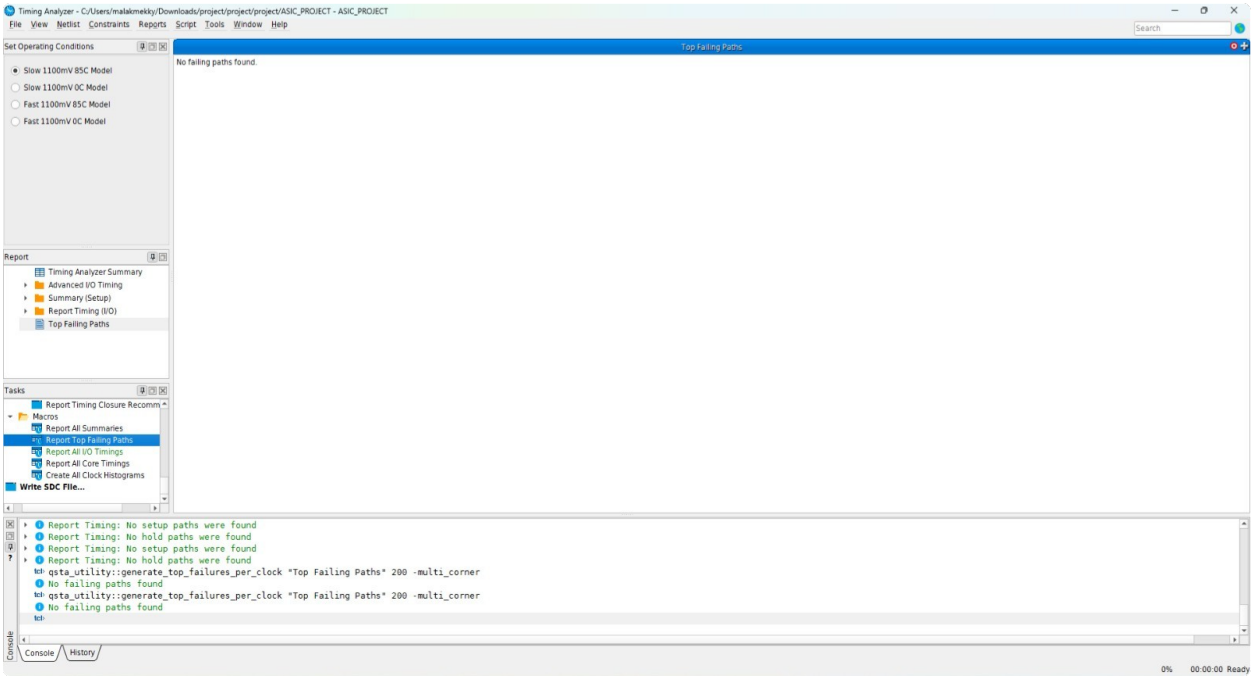
2. Timing Analysis

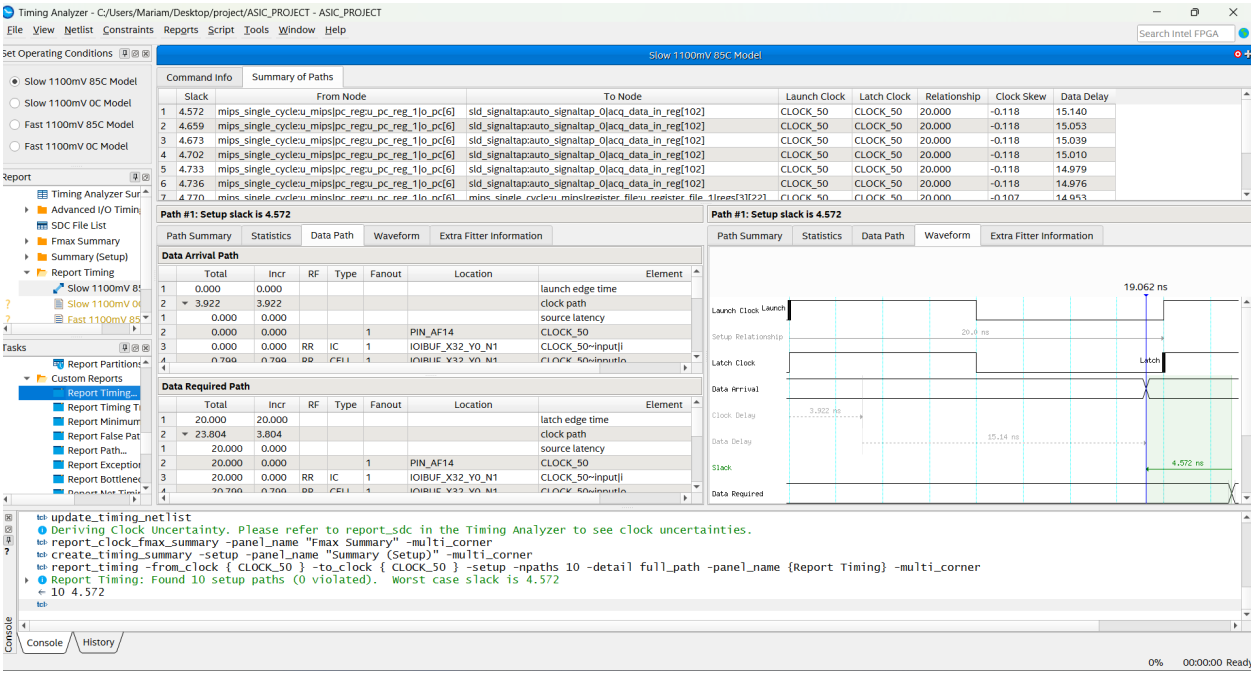


The processor's Fmax is 64.82 MHz at the slow 1100 mV 85C model, the worst case. This follows directly from the setup slack that we'll see below. The slowest path needs 20 - 4.572 = 15.428 ns, and 1/15.428 ns = 64.82 MHz.

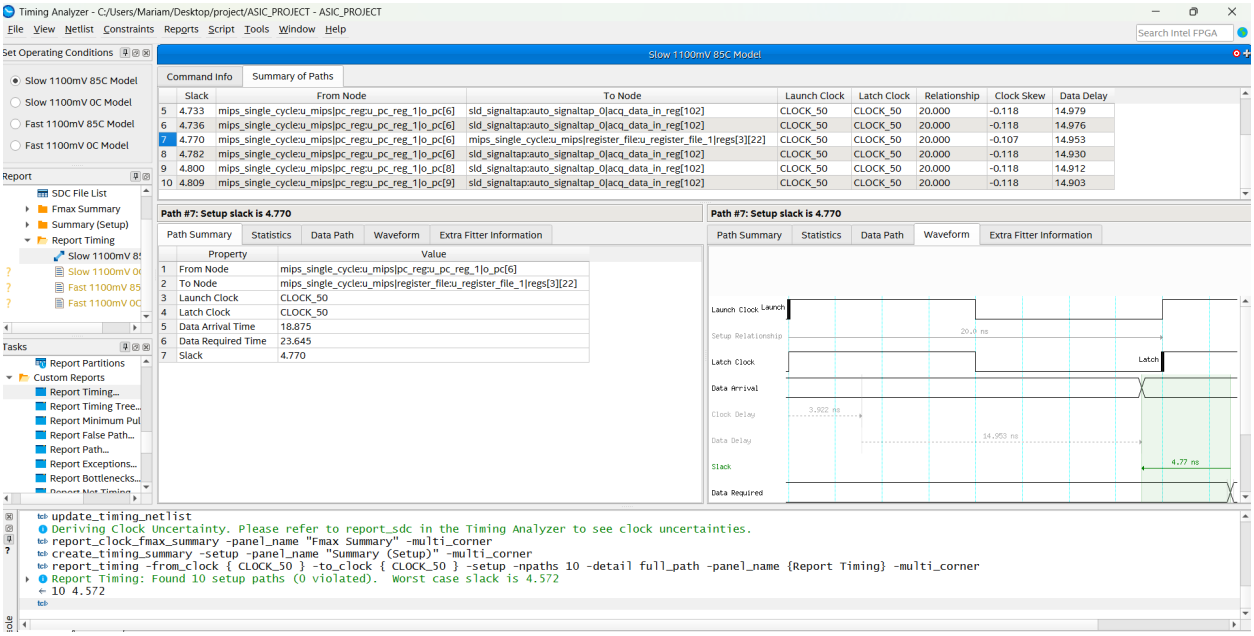


The setup summary shows a worst case slack of 4.572 with a total negative slack of 0 so no path fails.





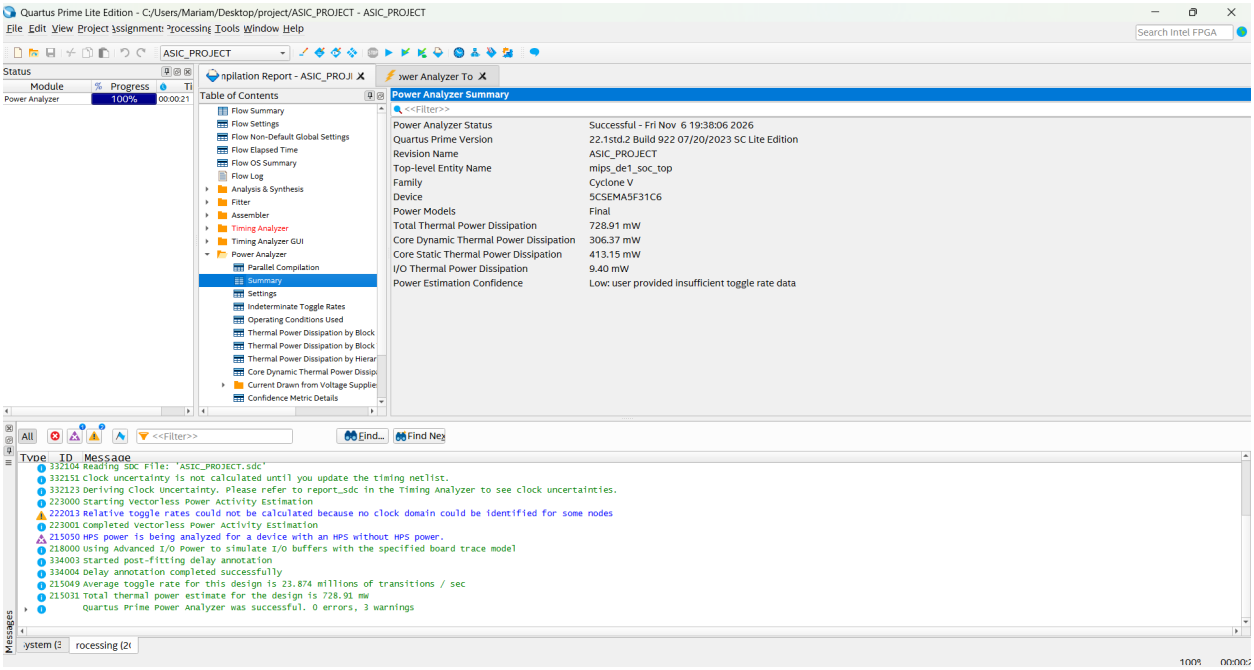
The ten slowest paths all start at the program counter which is expected. It's a single cycle processor so everything depends on the instruction the PC selects in a cycle. The worst path has a data delay of 15.140 ns after a clock delay of 3.922 ns, and arrives 4.572 ns before it's needed.



Path 7 is the slowest path that stays entirely inside the processor. It runs from the PC register to bit 22 of R3 in the register file, with a data delay of 14.953 ns and slack 4.770 ns.

Every path ending at the register file passes through the whole datapath in one cycle. The PC selects an instruction from the instruction memory, it sets the register addresses, immediate and control signals. The register file and ALU get a result and the result has to settle at the register file's write input before the next rising edge. This is why single cycle architecture limits the Fmax.

3. Power Analysis

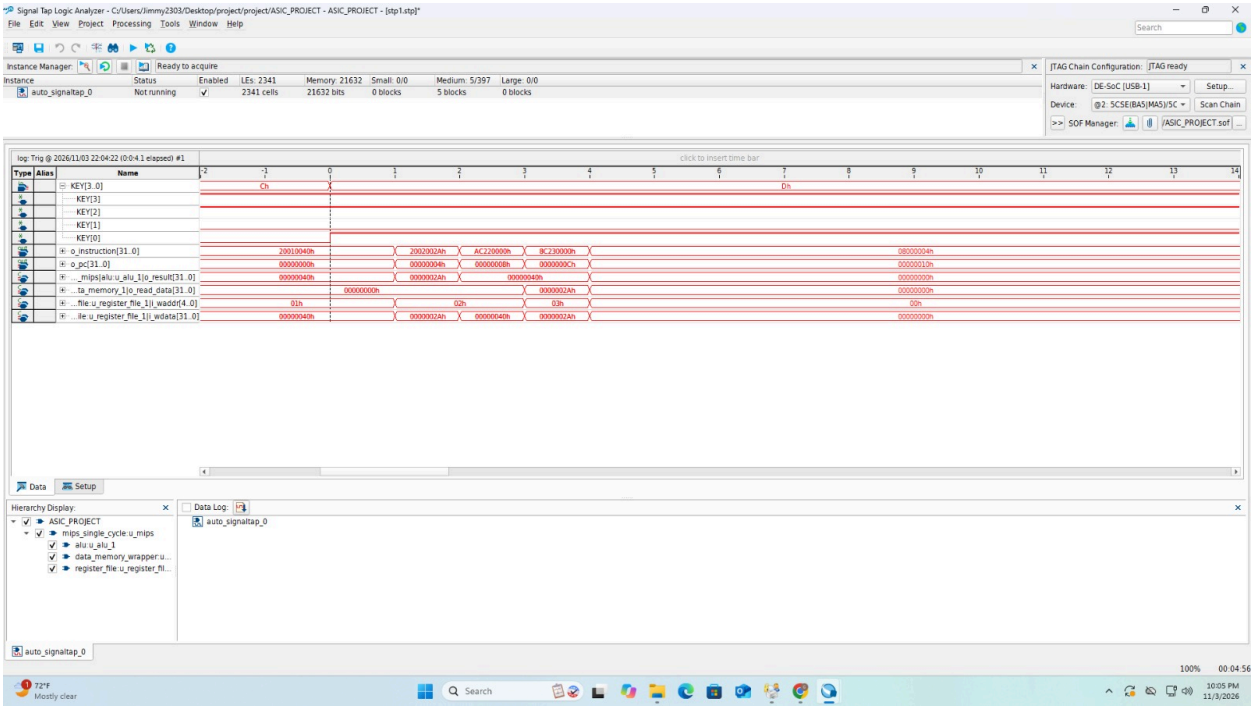


The Power Analyzer reports 728.91 mW in total. Also, 413.15 mW core static, 306.37 mW core dynamic and 9.40 mW I/O. Static power is 57% of the total. It's mostly the device's own leakage. Dynamic power is most likely the 10902 registers and the clock every cycle, also the 8192 flip flops.

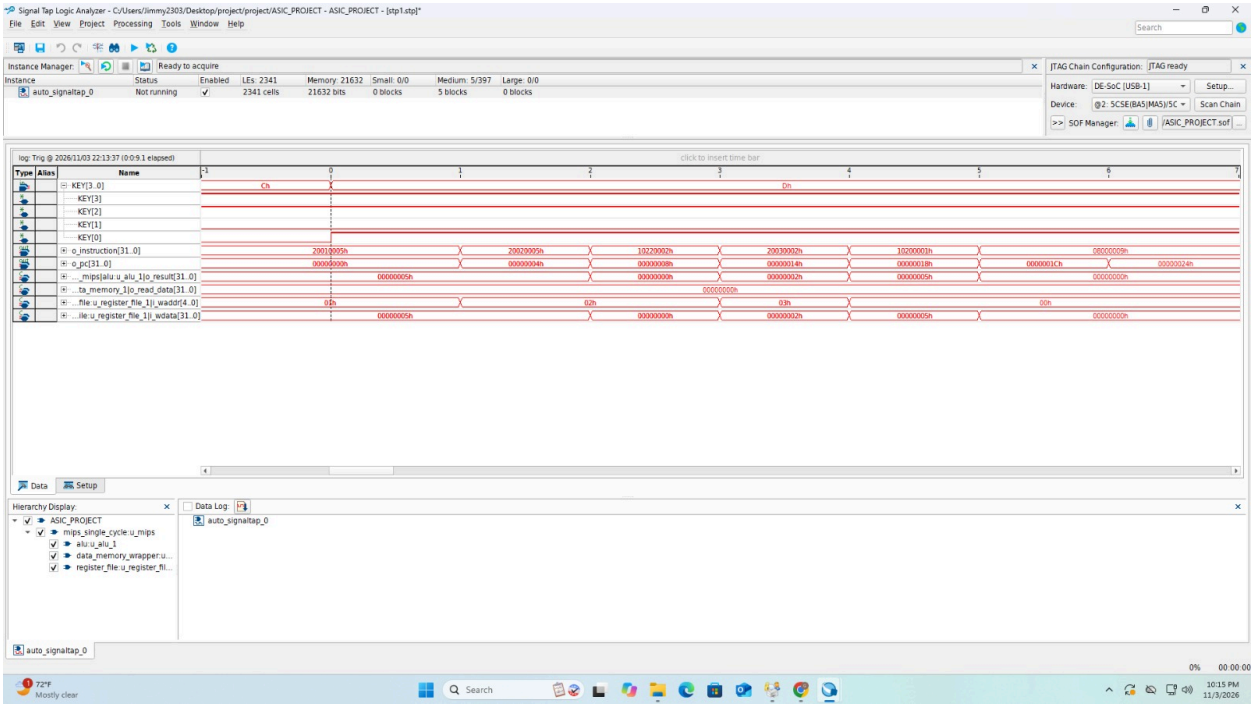
Energy per instruction = total power / clock frequency = 14.6 nJ

XIII. Board Demonstration

Test 2 Load & Store



Test 3: Branch & Jump



Test 4 Array Sum

